RotaryDryerDesigner

User Manual

MATLAB application for preliminary sizing
and axial simulation of direct-heated rotary dryers

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Contents

1. General description

RotaryDryerDesigner is a MATLAB application developed to support the preliminary sizing and axial simulation of direct-heated rotary dryers. The tool integrates within a single graphical environment the modules required for the definition of process conditions, material properties, kinetic fitting, model solution, and output analysis.

The purpose of the software is to facilitate the technical evaluation of drying cases through a structured workflow, enabling the relationship between input variables and the axial evolution of moisture content and temperature in the system, as well as key design parameters of the equipment. In addition, the application can be used as a supporting resource in engineering education, since it enables a guided exploration of the influence of operating conditions, solid properties, and kinetic parameters on system response.

2. Interface structure

RotaryDryerDesigner uses a MATLAB-based graphical user interface to integrate the main stages of case definition, calculation, and analysis for a rotary drying case within a single working environment. The application follows a logical workflow that allows the user to move from the initial process configuration to the visualization and interpretation of the computed variables.

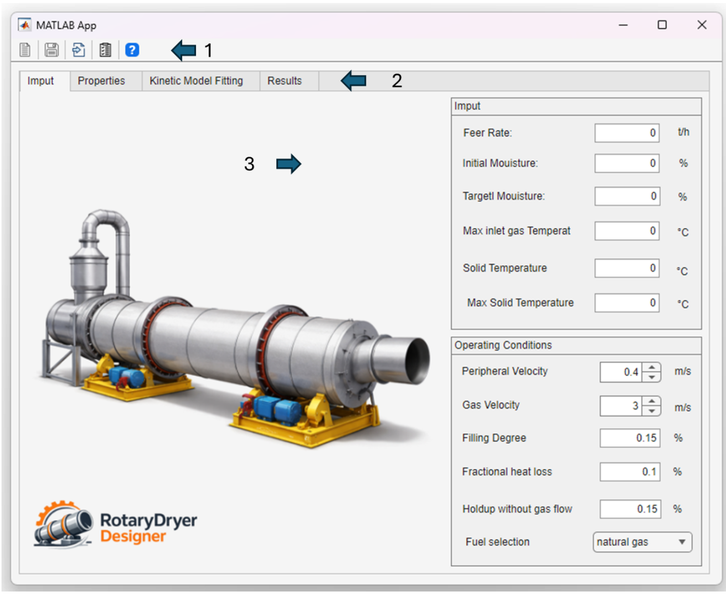

Figure 1: General working environment of RotaryDryerDesigner. (1) Top toolbar. (2) Navigation tab bar. (3) Main application workspace.

As shown in Figure 1, the main window contains the elements required for interaction with the application. The upper region includes a top toolbar with direct-access controls for creating a new project, saving the current project, importing a previously saved project, generating a report, and opening the help document. Below the toolbar, the navigation tab bar provides access to the main functional sections of the application, namely *Input*, *Properties*, *Kinetic Model Fitting*, and *Results*. The remaining area corresponds to the main workspace, where the controls, panels, tables, and plots associated with the active module are displayed.

From a functional perspective, this organization defines three complementary levels of interaction:

- **Global project actions**: accessed through the top toolbar.
- **Module navigation**: performed through the main application tabs.
- **Active workspace**: the area where input data are entered, calculations are executed, and outputs are displayed.

3. General workflow

The use of RotaryDryerDesigner follows an organized workflow structured around four main tabs. In general terms, the workflow is carried out as follows:

- In *Input*, the initial and operating conditions of the process are defined.
- In *Properties*, the physical and thermal properties required by the model are loaded, edited, or calculated.
- In *Kinetic Model Fitting*, the drying kinetic parameters are fitted from experimental data or entered manually when parameter values are already available for different temperatures.
- In *Results*, the final calculation settings are defined, the model is executed, and the graphical and tabulated outputs are analyzed.

3.1. Input tab

The *Input* tab is the starting point for configuring the study case, since it groups the initial parameters required to define the drying problem.

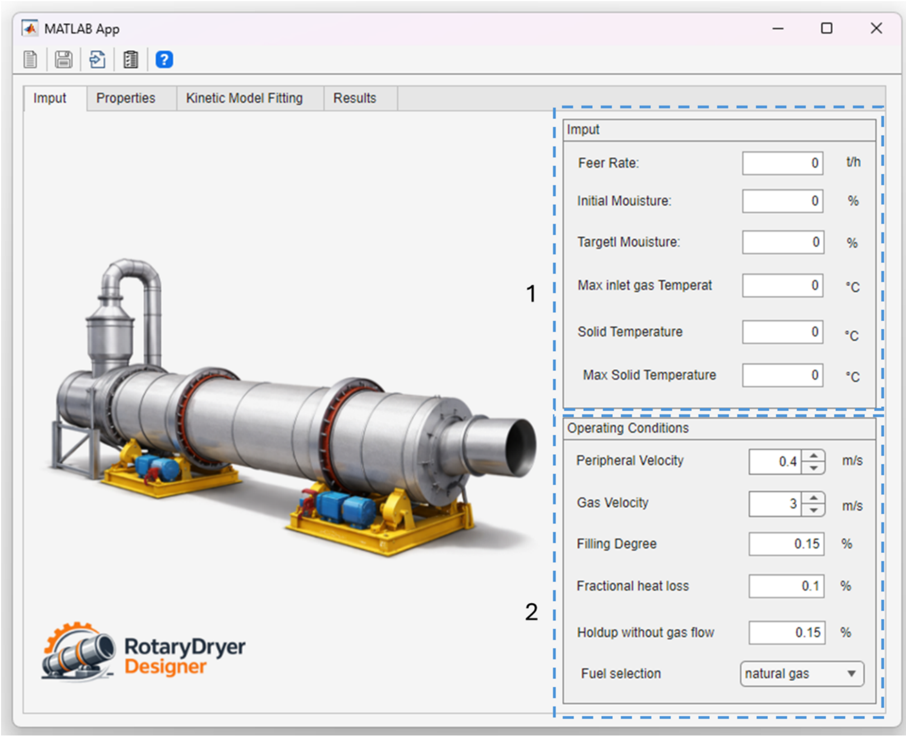

Figure 2: Input tab of RotaryDryerDesigner. (1) Process input variables block. (2) Operating conditions and fuel selection block.

As shown in Figure 2, this tab is organized into two main blocks:

- **Process input variables block**: includes feed rate, initial moisture content, target moisture content, maximum inlet gas temperature, initial solid temperature, and maximum allowable solid temperature. These parameters define the basic conditions from which the drying analysis is performed.
- **Operating conditions and fuel selection block**: contains variables such as peripheral velocity, gas velocity, filling degree, fractional heat loss, and holdup in the absence of gas flow. This block also includes fuel selection, which is used to complete the thermal configuration of the system.

3.2. Properties tab

The *Properties* tab groups the physical and thermal properties required to define the material and the gas streams involved in the process.

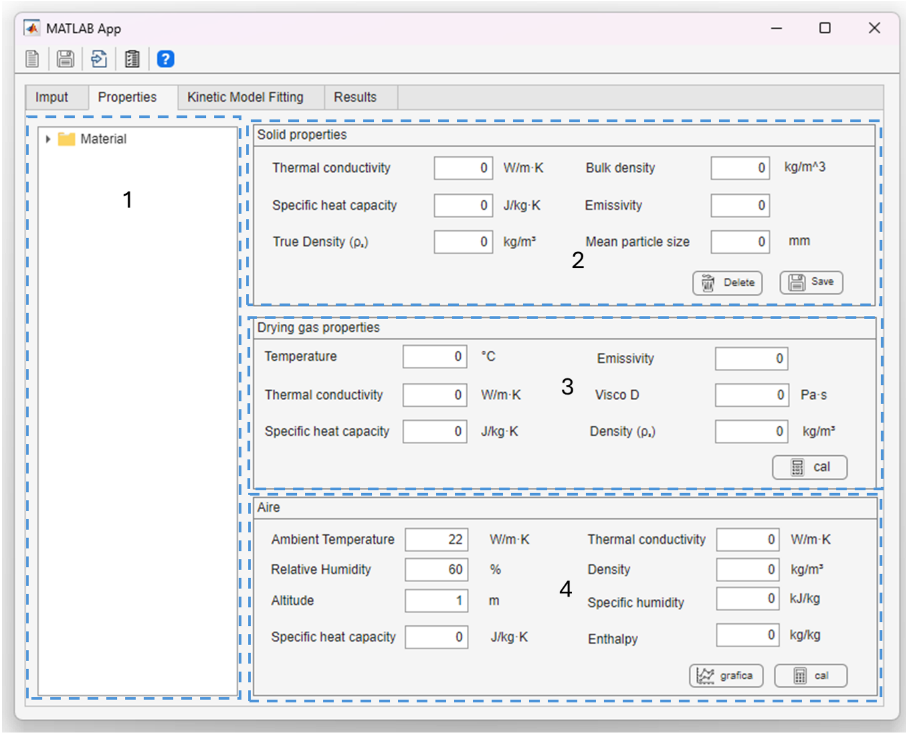

Figure 3: Properties tab of RotaryDryerDesigner. (1) Material selection panel. (2) Solid properties block. (3) Drying gas properties block. (4) Ambient air properties block.

As shown in Figure 3, this tab is divided into four functional blocks:

- **Material selection panel**: provides access to the material library available in the application. When a material is selected, its properties are automatically loaded into the corresponding solid properties block.
- **Solid properties block**: groups variables such as thermal conductivity, specific heat capacity, true density, bulk density, emissivity, and mean particle size. These values can be loaded from the library, edited manually, and saved again to the library or deleted as needed.
- **Drying gas properties block**: contains the variables required to represent the gaseous medium interacting with the solid. Based on the user-defined conditions, the application calculates properties such as temperature, thermal conductivity, specific heat capacity, emissivity, viscosity, and density.
- **Ambient air properties block**: groups variables such as temperature, relative humidity, and altitude, together with thermophysical properties calculated from these conditions. In addition, the application allows the air state to be displayed on the psychrometric chart.

3.3. Kinetic Model Fitting tab

The *Kinetic Model Fitting* tab allows drying kinetic models to be fitted from experimental data and subsequently used to estimate the Arrhenius parameters associated with their temperature dependence.

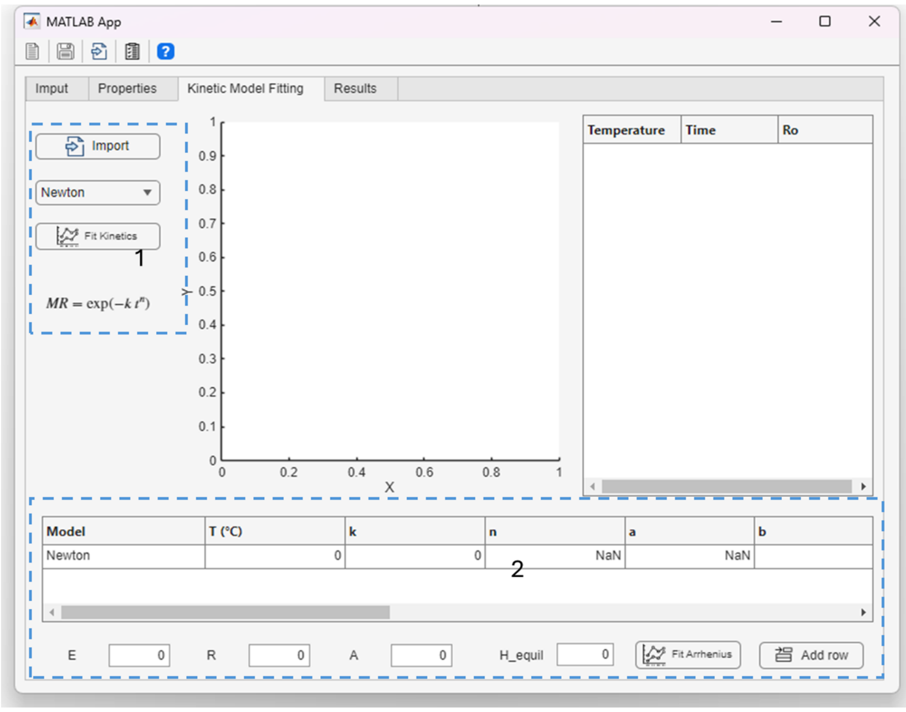

Figure 4: Kinetic Model Fitting tab of RotaryDryerDesigner. (1) Drying kinetic model import and fitting panel. (2) Kinetic parameters table and Arrhenius fitting block.

As shown in Figure 4, this tab is structured into two main blocks:

- **Drying kinetic model import and fitting panel**: allows experimental drying data to be imported, the kinetic model to be selected, and the fitting procedure to be executed. From experimental moisture-ratio series as a function of time at a given test temperature, the application estimates the constants of the selected model using the *Fit Kinetics* button.
- **Kinetic parameters table and Arrhenius fitting block**: automatically records the estimated constants for each temperature. This table can also be completed manually when the user already has experimentally reported parameters at different temperatures, and new rows can be added as required. Based on this information, the *Fit Arrhenius* button determines parameters such as activation energy and pre-exponential factor.

3.4. Results tab

The *Results* tab integrates the controls required to complete the design setup, configure the numerical solution, execute the axial model, and inspect the main outputs generated by the application. In this way, the tab functions as the final execution and analysis stage of the workflow.

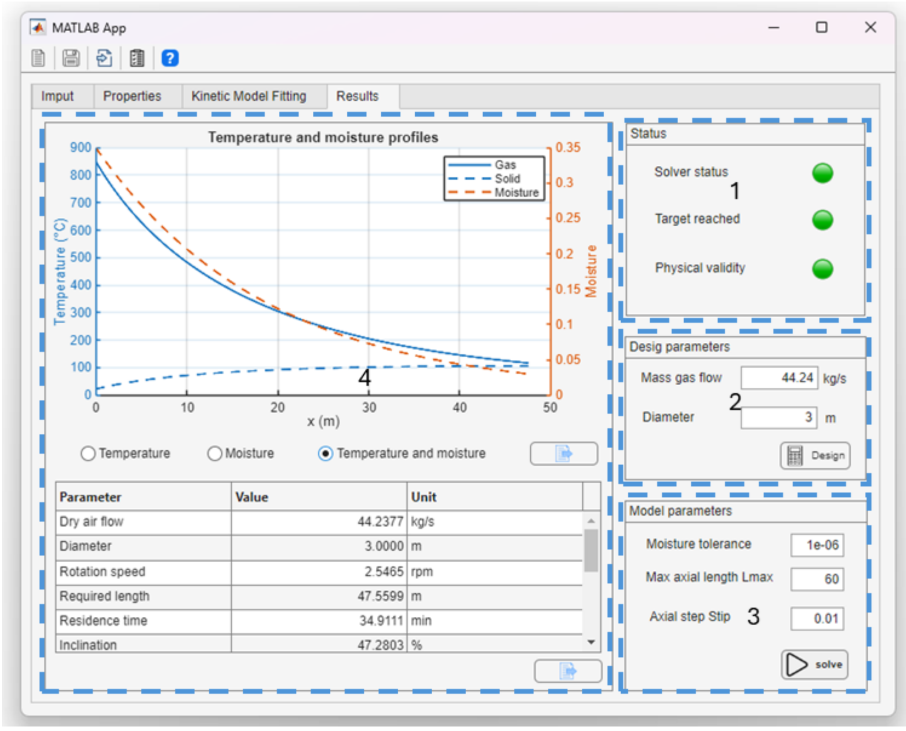

Figure 5: Results tab of RotaryDryerDesigner. (1) Status panel indicating solver condition, target satisfaction, and physical validity checks. (2) Design parameters block for preliminary dryer sizing. (3) Model parameters block for numerical configuration and solver execution. (4) Main visualization area for the calculated axial profiles and Summary results table with the main calculated outputs.

As shown in Figure 5, the *Results* tab is organized into six main functional areas:

- **Status panel**: displays basic diagnostic indicators associated with the current solution. These indicators inform whether the solver completed successfully, whether the target outlet condition was reached, and whether the calculated solution satisfies the implemented physical-validity checks.
- **Design parameters block**: allows the user to define the main variables required for preliminary dryer sizing, including the drying-gas mass flow rate and dryer diameter. The *Design* button uses these inputs to estimate the equipment dimensions required to satisfy the specified process conditions.
- **Model parameters block**: contains the numerical settings used during the axial simulation, including the moisture tolerance, maximum axial length, and axial

step size. This block also includes the *solve* button, which launches the model solution using the current design and operating conditions.

- **Main visualization area**: presents the calculated axial profiles of the process variables. In the configuration shown, the interface displays gas temperature, solid temperature, and moisture evolution along the dryer length, allowing direct inspection of the predicted thermal and drying behavior.
- **Display controls**: provide options to select the type of graphical representation to be shown, including temperature only, moisture only, or the combined temperature-and-moisture view. This area also includes the plot-export control for saving the displayed figure.
- **Summary results table**: reports the main calculated outputs in a structured format with parameter name, numerical value, and corresponding unit. Typical outputs include gas flow rate, dryer diameter, rotation speed, required length, residence time, and inclination, thereby supporting the interpretation and documentation of the simulated case.

4. Warnings and recommendations

For proper use of the application, the following aspects should be considered:

- Verify that all input variables are entered in the units indicated by the interface.
- Check that the properties of the selected material correspond to the solid of interest before executing the model.
- Confirm that the experimental data used for kinetic fitting are associated with the correct test temperature.
- Select numerical parameters consistent with the desired model resolution before running the solution.
- Analyze the graphical and tabulated outputs jointly in order to support their technical interpretation.